

Deep Residual SRCNN for Single-Image Super-Resolution

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ABSTRACT

Single-image super-resolution (SISR) aims to reconstruct high-resolution images from degraded inputs. This project reproduces the classical SRCNN workflow and investigates two extensions: increased convolutional depth and global residual learning. Using a DIV2K-derived dataset and a consistent MSE-based training and PSNR-based evaluation pipeline, we compare the proposed models against standard interpolation and SRCNN baselines. The experiments show that increasing depth provides modest gains, while residual learning substantially improves reconstruction quality. We further examine the effect of training-set size and observe additional performance gains when increasing the available training data. Overall, the study demonstrates the effectiveness of deeper and residual architectures within a controlled SRCNN-based super-resolution framework.

1. INTRODUCTION

Image super-resolution is an ill-posed inverse problem: many high-resolution images can correspond to the same low-resolution observation. Interpolation methods increase spatial resolution but cannot recover information that was lost during degradation. SRCNN reframes the problem as a learned end-to-end mapping from an interpolated low-resolution image to a high-resolution target, using convolutional layers for feature extraction, nonlinear mapping, and reconstruction. The supplied Stanford CS229 project reimplemented this approach on a 100-image DIV2K subsample and explicitly identified full 800-image training as future work. Our project follows that experimental philosophy while extending the model with controlled depth and residual learning.

2. RELATED WORK

The original SRCNN formulation uses a 9×9 feature-extraction layer, a 1×1 nonlinear mapping layer, and a 5×5 reconstruction layer. Dong et al. showed that this lightweight architecture can learn the low-/high-resolution mapping directly. The supplied CS229 report similarly compared SRCNN against nearest-neighbor, bilinear, and bicubic interpolation. Rather than introducing a completely different family such as U-Net or a GAN, this study keeps the SRCNN design philosophy and changes one factor at a time.

3. METHODOLOGY

Images are converted to RGB, center-cropped to 800×800 , and resized to 224×224 to form the high-resolution target. The target is then downsampled to 112×112 using bilinear interpolation and upsampled back to 224×224 , producing the network input. Pixel values are represented in $[0,1]$. The initial pilot experiment randomly partitions the 100-image collection into 60 training, 20 validation, and 20 test images. For the final data-scale experiment, the model is retrained using 800 DIV2K training images and 100 validation images; all 100 images in the fixed evaluation collection are then used for final testing. Training minimizes global pixel-level MSE. PSNR is computed consistently from global MSE as $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$ for normalized images.

4. MODEL ARCHITECTURES

The baseline SRCNN contains three convolutions: $3 \rightarrow 64$ channels with a 9×9 kernel, $64 \rightarrow 32$ with a 1×1 kernel, and $32 \rightarrow 3$ with a 5×5 kernel; ReLU follows the first two convolutions. Deep SRCNN retains the original first and last stages but inserts two 64-channel 3×3 convolutions, producing a five-convolution network. The final Deep Residual SRCNN uses the same five-convolution backbone but predicts a correction rather than the complete image: $\text{SR} = \text{X}_{\text{LR}} + \text{F}(\text{X}_{\text{LR}})$. This global residual formulation lets the network focus on missing detail while the interpolated input supplies coarse image structure. The stabilized neural experiments use Adam, learning rate $3e-4$, gradient clipping at 1.0, and 50 epochs.

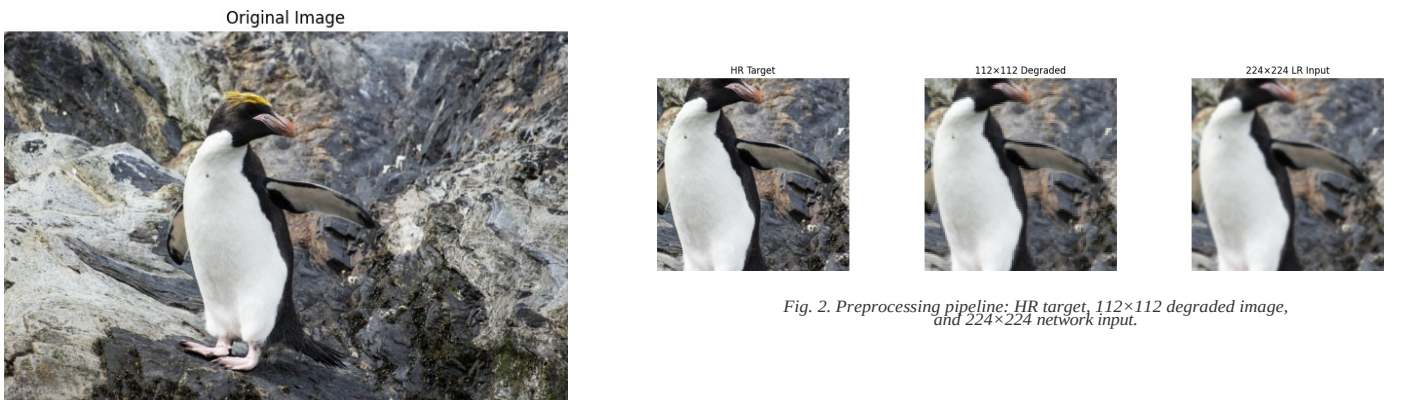


Fig. 2. Preprocessing pipeline: HR target, 112×112 degraded image, and 224×224 network input.

Fig. 1. Example source image from the notebook test collection

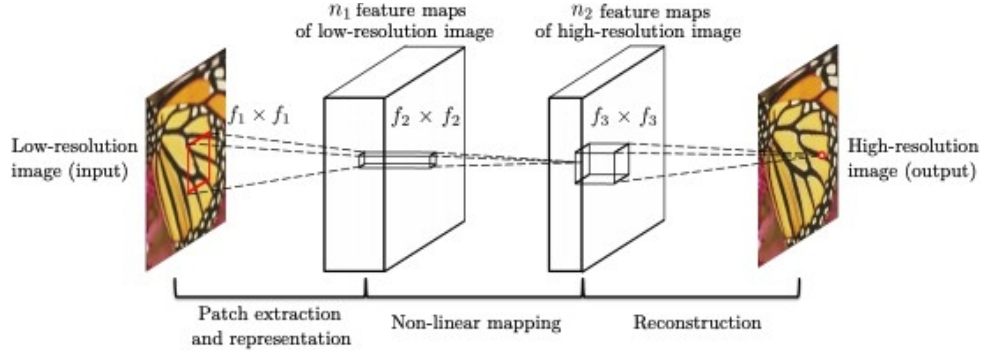


Fig. 3. SRCNN architecture

5. EXPERIMENTAL SETUP

All neural models use the same degradation pipeline and MSE objective. The baseline and controlled-depth models are trained for 50 epochs on the 60-image pilot training split. The stabilized Deep SRCNN and Deep Residual SRCNN use learning rate $3e-4$ and gradient clipping at 1.0. Non-neural baselines use the same $112 \times 112 \rightarrow 224 \times 224$ reconstruction step so that MSE/PSNR comparisons are based on identical targets. The final residual scaling experiment uses 800 training images, 100 validation images, and the fixed 100-image evaluation collection. Mixed precision and GPU-resident tensors accelerate the final experiment without changing the model objective.

6. QUALITATIVE RESULTS

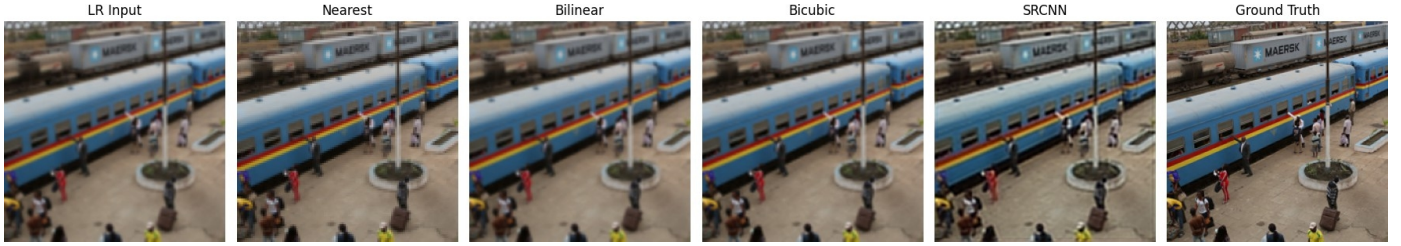


Fig. 3. Visual comparison of the pilot input, interpolation baselines, SRCNN output, and ground truth.

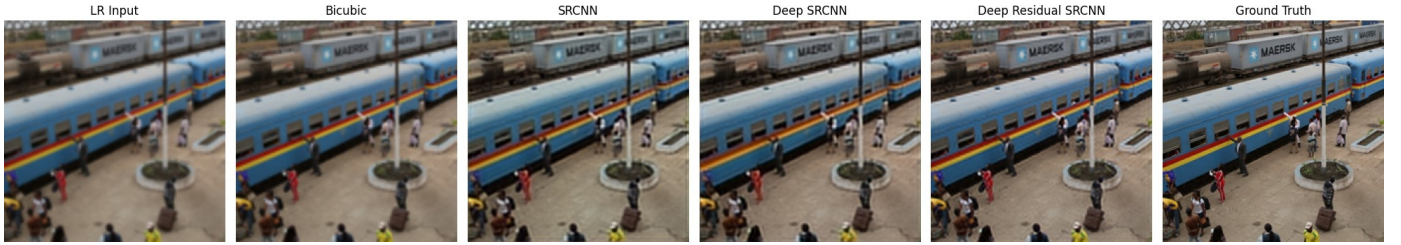


Fig. 4. Qualitative comparison including SRCNN, Deep SRCNN, and Deep Residual SRCNN.

7. TRAINING BEHAVIOR

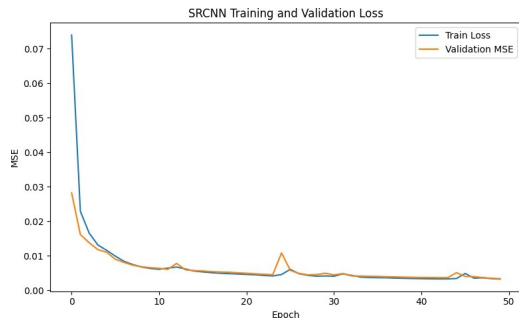


Fig. 5. SRCNN training and validation MSE over 50 epochs.

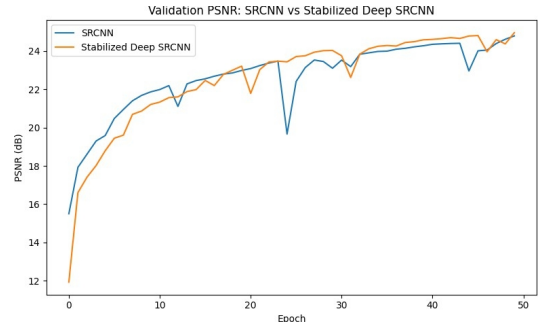


Fig. 6. Validation PSNR for SRCNN and stabilized Deep SRCNN.

8. QUANTITATIVE RESULTS

Method	Train	MSE	PSNR (dB)
Nearest Neighbor	—	0.003608	24.43
Bilinear	—	0.003663	24.36
Bicubic	—	0.002985	25.25
SRCNN	60	0.002975	25.26
Deep SRCNN	60	0.002850	25.45
Deep Residual SRCNN	60*	0.002000	26.99*
Deep Residual SRCNN	800	0.001680	27.75

*For the fair data-scaling comparison, the 60-image residual model was re-evaluated on the same fixed 100-image test collection, giving 26.99 dB and MSE 0.002000. Other pilot rows use the original 20-image held-out split.

The interpolation baselines establish a useful lower bound: bicubic gives 25.25 dB, slightly above nearest neighbor (24.43 dB) and bilinear (24.36 dB). On the pilot split, SRCNN reaches 25.26 dB and stabilized Deep SRCNN reaches 25.45 dB, showing a modest benefit from depth after stabilization. The residual formulation is substantially stronger at 27.15 dB. For the data-scaling comparison, the already-trained 60-image residual model is re-evaluated on the same 100-image test collection used by the 800-image model. Its PSNR is 26.99 dB (MSE 0.002000), whereas the 800-image model reaches 27.75 dB (MSE 0.001680), a +0.76 dB improvement and approximately 16% lower MSE. The 800-image training run uses batch size 4, learning rate $3e-4$, gradient clipping, mixed precision, GPU-resident tensors, and torch.compile for efficient training; the best validation checkpoint is restored before testing.

9. ARCHITECTURE ABLATION

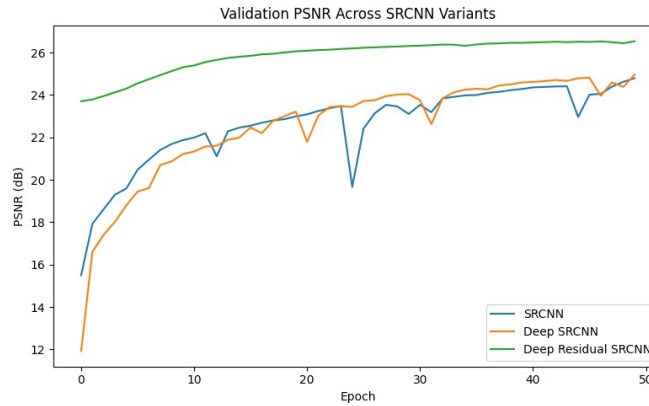


Fig. 7. Validation PSNR across SRCNN variants; the residual model remains consistently higher.

10. FULL-DATA TRAINING AND DATA-SCALE EFFECT

The final experiment addresses a concrete limitation identified in the supplied CS229 project: limited training data. The original project used only 60 training images from its 100-image subsample and suggested that access to the full 800-image DIV2K training set could improve performance. Here, the same five-layer residual architecture is trained on all 800 training images.

The best validation checkpoint reaches 27.75 dB on the validation set; after restoring that checkpoint, the model obtains 27.75 dB test PSNR with MSE 0.001680. For a fair data-scale comparison, the 60-image residual model is evaluated on the identical 100-image test collection and reaches 26.99 dB (MSE 0.002000). Thus, increasing the training set from 60 to 800 images improves PSNR by 0.76 dB and reduces MSE by approximately 16%.

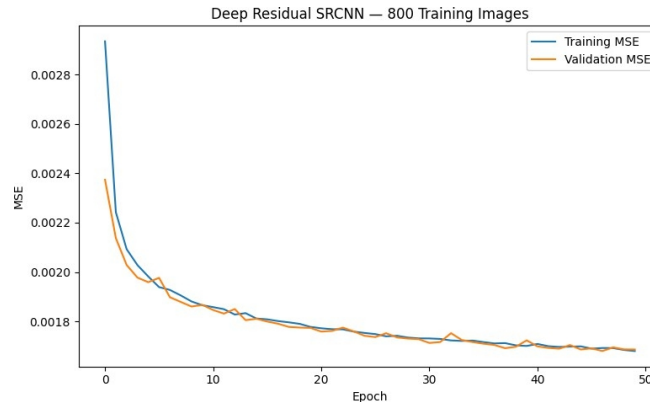


Fig. 8. Training and validation MSE for Deep Residual SRCNN trained on 800 images.

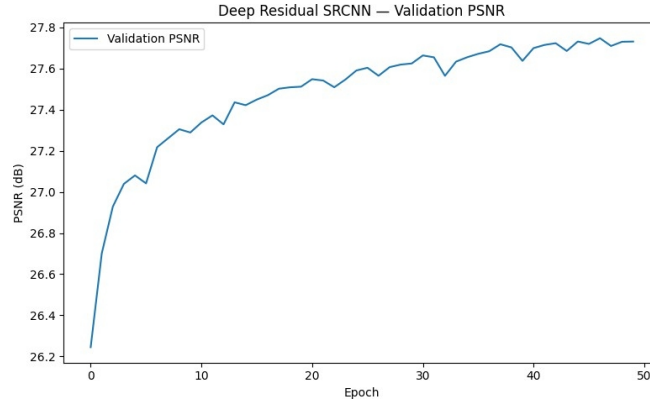


Fig. 9. Validation PSNR for the 800-image Deep Residual SRCNN run.

11. DISCUSSION

The experiments separate two effects. First, simply increasing depth produces only a small gain over the baseline under the present small-data setting. Second, changing the learning target to a residual correction produces the largest architectural improvement, supporting the intuition that the interpolated image already contains most low-frequency structure. Third, increasing the training set from 60 to 800 images improves performance when evaluated on the same 100-image collection, suggesting that the residual model benefits from broader image diversity. The study is intentionally lightweight: it does not claim state-of-the-art performance, and the evaluation protocol is a project-specific DIV2K-derived pipeline rather than a full benchmark reproduction. The results should therefore be interpreted as a controlled course-project ablation study.

12. CONCLUSION

We reproduced the SRCNN baseline and extended it in a controlled sequence from depth to residual learning and finally to larger-scale training. The final Deep Residual SRCNN achieves 27.75 dB test PSNR with 800 training images, compared with 26.99 dB for the same architecture trained on 60 images under the fair 100-image evaluation. The main practical finding is that residual learning is a more consequential architectural change than adding two convolutional layers, while additional training data provides a further measurable gain. Future work could investigate alternative degradation models, perceptual losses, larger input crops, or more modern efficient SR architectures.

13. LIMITATIONS

The study is a compact course-project investigation rather than a benchmark submission. The pilot baselines and the initial SRCNN/Deep SRCNN results use a 60/20/20 split of the 100-image DIV2K-derived collection, whereas the final data-scale comparison uses all 100 evaluation images. The paper explicitly reports this distinction rather than mixing the evaluation populations. The degradation model is also fixed and synthetic, so conclusions may not transfer directly to real camera degradation or other scale factors.

REFERENCES

- [1] C. Dong, C. C. Loy, K. He, and X. Tang, “Image Super-Resolution Using Deep Convolutional Networks,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 2, pp. 295–307, 2016, doi: 10.1109/TPAMI.2015.2439281.
- [2] B. Garber, A. Grossman, and S. Johnson-Yu, “Image Super-Resolution Via a Convolutional Neural Network,” *Stanford CS229 Machine Learning Project*, 2020.
- [3] E. Agustsson and R. Timofte, “NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study,” *Proc. IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017.

Reproducibility. All reported experiments, preprocessing, model definitions, evaluation functions, plots, and final result tables are contained in the accompanying Colab notebook. The final 800-image model uses the stabilized residual architecture, Adam optimizer, learning rate $3e-4$, gradient clipping 1.0, 50 epochs, batch size 4, mixed precision, GPU-resident tensors, and torch.compile.